

KVM/QEMU Hypervisor Lab

Day 2 Lab Journal — DC01 Deployment & AD DS Configuration

Date: June 23, 2026 | Engineer: Demi | Host OS: Ubuntu 24.04.3 LTS

Placement: Navy QUORRA — SIWES Industrial Training

1. Day 2 Objective

Deploy Windows Server 2022 as a KVM virtual machine (DC01), install VirtIO drivers for disk and network, configure a static IP, promote the server to an Active Directory Domain Controller for lab.local, and configure DHCP. Take a clean baseline snapshot on completion.

2. Phase 5 — DC01 VM Creation

2.1 Disk Image Creation

Created an 80GB qcow2 disk image on the SSD-backed storage pool:

```
sudo qemu-img create -f qcow2 /var/lib/libvirt/images/dc01.qcow2 80G
```

Note: First attempt failed due to a typo — path was written as /var/lib/libvirt/images (double r). Corrected and re-run successfully.

2.2 virt-install — Command Not Found

Attempted to launch the VM with virt-install. Received 'command not found' despite the package being installed. Root cause: sudo drops the user PATH, so virt-install at /usr/bin/virt-install was not accessible.

Resolution:

```
sudo env "PATH=$PATH" virt-install ...
```

Passing the current PATH explicitly to sudo resolved the issue.

2.3 Graphics Argument Error

First virt-install attempt failed with:

```
Error: unrecognized arguments: Listen=0.0.0.0, port=5910
```

Root cause: Spaces in the --graphics argument were being parsed as separate tokens. Simplified the graphics flag to remove explicit listen/port values and let libvirt assign defaults.

```
--graphics vnc
```

2.4 VM Successfully Created

Final working virt-install command (single line, no backslash continuation):

```
sudo env "PATH=$PATH" virt-install --name DC01 --ram 8192 --vcpus 2 --cpu host --disk
```

```
path=/var/lib/libvirt/images/dc01.qcow2,format=qcow2,bus=virtio --cdrom
/mnt/data/isos/Windows_Server_2022.iso --os-variant win2k22 --network
network=labnet,model=virtio --graphics vnc --noautoconsole --boot
cdrom,hd
```

Output confirmed: Domain is still running. Installation may be in progress.

3. VNC Console Access

3.1 Why a Second Machine Was Needed

The Ubuntu hypervisor runs as a headless server with no desktop environment installed. When DC01 boots, its display output is served via VNC but has no local screen to render on. A second machine was required purely as a viewing terminal — DC01 runs 100% on the hypervisor. The second machine is temporary scaffolding, not a dedicated host for Windows Server.

3.2 VNC Port Discovery

```
virsh vncdisplay DC01
```

Output: 127.0.0.1:10 — meaning VNC was bound to localhost on port 5910 (display :10). External direct access was blocked.

3.3 SSH Tunnel Setup

Hypervisor IP confirmed as 192.168.1.29 via `ip addr show br0` (enp2s0 was bridged to br0). SSH tunnel command run from Windows laptop:

```
ssh -L 5910:127.0.0.1:5910 username@<hypervisorip>
```

RealVNC Viewer installed on Windows laptop. Connected to localhost:5910 — Windows Server 2022 installer appeared successfully.

4. Windows Server 2022 Installation

4.1 Installer Options Selected

- Language: English
- Installation type: Custom (Advanced)
- Edition: Windows Server 2022 Standard Evaluation (Desktop Experience)

4.2 VirtIO Disk Driver — No Disk Visible

On the 'Where do you want to install Windows' screen, no drives appeared. Windows does not include VirtIO drivers natively and could not see the qcow2 disk attached via the virtio bus.

First Attempt — attach-disk Command:

```
virsh attach-disk DC01 /mnt/data/isos/virtio-win.iso sdb --type cdrom
--mode readonly
```

Failed with: No more available PCI slots. DC01 was created with all PCI slots occupied.

Resolution — Edit VM XML Directly:

Shut down DC01 and edited the XML configuration to add a second SATA cdrom entry:

```
sudo env "PATH=$PATH" virsh edit DC01
```

Added the following block after the existing Windows ISO cdrom entry:

```
<disk type='file' device='cdrom'>
  <driver name='qemu' type='raw' />
  <source file='/mnt/data/isos/virtio-win.iso' />
  <target dev='sdb' bus='sata' />
  <readonly />
  <address type='drive' controller='0' bus='0' target='0' unit='1' />
</disk>
```

XML Validation Error:

First save attempt failed with: Opening and ending tag mismatch: disk line 55 devices. A stray self-closing <disk/> tag was accidentally inserted. Located and deleted it, re-saved successfully.

Driver Installation:

Back in the Windows installer, clicked Load Driver → Browse → navigated to E:\viorstor\2k22\amd64. Selected Red Hat VirtIO SCSI controller. 80GB disk appeared and installation proceeded.

4.3 Installation Completed

Windows Server 2022 installed successfully. VM rebooted automatically into the out-of-box setup screen. Administrator password set.

5. Phase 6 — Post-Install Configuration

5.1 VirtIO Network Driver

After logging in, Get-NetAdapter returned no results — only the Microsoft Kernel Debug Network Adapter was visible. The VirtIO network driver was not installed.

Resolution:

Opened Device Manager → located unknown network device → Update Driver → Browse to E:\NetKVM\2k22\amd64. Windows installed the Red Hat VirtIO Ethernet Adapter successfully. Get-NetAdapter then showed interface name: Ethernet

5.2 Static IP Configuration

```
New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.10.10
-PrefixLength 24 -DefaultGateway 192.168.10.1
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses
127.0.0.1,8.8.8.8
```

Both commands completed without errors.

5.3 Hostname

```
Rename-Computer -NewName "name" -Restart
```

Server rebooted and came back up as DC01.

5.4 Role Installation

```
Install-WindowsFeature -Name AD-Domain-Services, DNS, DHCP  
-IncludeManagementTools
```

Output: Success True — all three roles installed successfully.

5.5 Domain Controller Promotion

```
Install-ADDSForest -DomainName "lab.local" -DomainNetbiosName "LAB"  
-SafeModeAdministratorPassword (...) -InstallDns -Force
```

PowerShell cursor blinked for approximately 5-10 minutes while AD DS database, DNS zones, and SYSVOL were configured. Server rebooted automatically on completion.

Post-reboot login confirmed with LAB\Administrator credentials.

5.6 DHCP Configuration

```
Add-DhcpServerInDC -DnsName "DC01.lab.local" -IPAddress 192.168.10.10  
Add-DhcpServerv4Scope -Name "LabScope" -StartRange 192.168.10.50  
-EndRange 192.168.10.200 -SubnetMask 255.255.255.0  
Set-DhcpServerv4OptionValue -ScopeId 192.168.10.0 -Router 192.168.10.1  
-DnsServer 192.168.10.10  
Restart-Service dhcpserver
```

Verified with Get-DhcpServerv4Scope and Get-DhcpServerInDC — both returned correct configuration.

6. Baseline Snapshot

```
virsh snapshot-create-as DC01 "baseline-ad-configured" --description  
"Clean AD DS DNS DHCP after promotion" --atomic
```

Output: Domain snapshot baseline-ad-configured created

This snapshot serves as a clean restore point before any further configuration, GPO changes, or user creation. If DC01 is ever broken during lab exercises, it can be rolled back to this state instantly.

7. Errors & Resolutions Summary

Error	Root Cause	Resolution
qemu-img: path typo libvirt	Double r typo in /var/lib/libvirt path	Re-ran with correct path
virt-install: command not found	sudo drops user PATH, binary at /usr/bin not found	sudo env "PATH=\$PATH" virt-install

Graphics argument unrecognized	Spaces in --graphics flag parsed as separate tokens	Simplified to --graphics vnc
No disk visible in Windows installer	VirtIO disk driver not included in Windows by default	Loaded viostor driver from virtio-win ISO via Load Driver
virsh attach-disk: No more PCI slots	All PCI slots were used at VM creation time	Edited VM XML directly to add second SATA cdrom
XML tag mismatch error on save	Stray <disk/> self-closing tag accidentally inserted	Located and deleted the rogue tag in nano
Get-NetAdapter: no results	VirtIO network driver not installed in Windows	Installed netkvm driver via Device Manager from virtio ISO
New-NetIPAddress: Element not found	Interface name was not 'Ethernet' initially (no driver)	Fixed after installing VirtIO network driver

8. End of Day 2 Status

Milestone	Status
DC01 VM created (80GB qcow2, 8GB RAM, 2 vCPU)	✓ Complete
Windows Server 2022 installed (Desktop Experience)	✓ Complete
VirtIO disk driver installed (viostor)	✓ Complete
VirtIO network driver installed (netkvm)	✓ Complete
Static IP set — 192.168.10.10/24	✓ Complete
Hostname set to DC01	✓ Complete
AD DS, DNS, DHCP roles installed	✓ Complete
lab.local domain created	✓ Complete
DC01 promoted to Domain Controller	✓ Complete
DHCP scope configured (192.168.10.50-200)	✓ Complete
Baseline snapshot taken	✓ Complete
VM2 — Linux server joined to domain	⌚ Day 3
VM3 — Windows client for GPO testing	⌚ Day 3
Group Policy configuration	⌚ Day 3

Day 3 — Next Steps

- Deploy VM2 — Rocky Linux or Ubuntu Server joined to lab.local
- Deploy VM3 — Windows 10/11 client for GPO and domain login testing
- Create Organisational Units (OUs) in Active Directory

- Configure Group Policy Objects (GPOs)
- Create test domain user accounts
- Copy DC01 snapshot to /mnt/backup/vm-snapshots on HDD2